TOPS Technologies

Interview Questions — SQL for Data Analytics

Total Questions: 56

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1. What is a database, and how is it different from a spreadsheet?

> **Hint:** Think about how data is structured, accessed, and managed in each.

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2. Explain what RDBMS stands for and why it is important in data analytics.

> **Hint:** Focus on how data is stored in tables and relationships.

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3. Name two popular SQL database engines and mention one key difference between them.

> **Hint:** Consider MySQL, PostgreSQL, or Oracle.

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4. After installing MySQL, how would you create a new database called 'analytics_db'?

> **Hint:** Recall the SQL command for database creation.

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5. What does the SELECT statement do in SQL?

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6. How would you select all columns from a table named 'employees'?

> **Hint:** There's a shortcut symbol for all columns.

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7. How can you rename a column in your SELECT query output using SQL?

> **Hint:** Look for a keyword that allows aliasing.

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8. Write a SQL query to select the names and salaries of employees from the 'employees' table where the salary is greater than 50,000 and the department is 'IT'.

> **Hint:** Combine WHERE with AND.

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9. What is the difference between the = and > operators in SQL?

> **Hint:** One checks for equality, the other for inequality.

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10. How does the OR logical operator work in a WHERE clause?

> **Hint:** It returns rows if at least one condition is true.

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11. Write a query to find all customers whose names start with the letter 'M'.

> **Hint:** Which operator is used for pattern matching?

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12. What does the BETWEEN operator do in SQL?

> **Hint:** It's used to filter results within a range.

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13. How would you select products with IDs in the set (101, 104, 108)?

> **Hint:** There's a specific operator for matching a list of values.

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14. What is the purpose of the DISTINCT keyword in a SELECT statement?

> **Hint:** It helps remove duplicate values.

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15. How do you sort query results in descending order based on a column named 'revenue'?

> **Hint:** Which keyword controls sorting direction?

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16. Write a query to get the top 10 highest selling products from a 'sales' table.

> **Hint:** Use ORDER BY and LIMIT together.

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17. Name three aggregate functions in SQL and briefly describe what they do.

> **Hint:** Think about SUM, COUNT, AVG, etc.

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18. Write a SQL query to calculate the average salary from the 'employees' table.

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19. How would you round a numeric result to 2 decimal places in SQL?

> **Hint:** There's a function specifically for rounding.

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20. What is the difference between WHERE and HAVING clauses in SQL?

> **Hint:** Consider when aggregation has happened.

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21. Write a query to show total revenue per region, but only for regions where revenue is greater than 1,000,000.

> **Hint:** Combine GROUP BY with HAVING.

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22. Explain the difference between INNER JOIN, LEFT JOIN, and RIGHT JOIN.

> **Hint:** Focus on which rows are included from each table.

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23. Write a query to join 'orders' and 'customers' tables and show each order along with the customer's segment.

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24. What is a primary key and why is it important in relational databases?

> **Hint:** Think about uniqueness and table relationships.

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25. When would you use a FULL OUTER JOIN instead of an INNER JOIN?

> **Hint:** Consider what happens to unmatched rows.

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26. Explain what a self join is and give a scenario where it might be useful.

> **Hint:** Think about hierarchical relationships, like employees and managers.

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27. What is a cross join and what kind of result does it produce?

> **Hint:** Think about the number of rows in the result.

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28. What is the difference between UNION and UNION ALL?

> **Hint:** Consider how duplicates are handled.

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29. Write a query to combine results from 'online_sales' and 'offline_sales' tables, showing all records including duplicates.

> **Hint:** Which operator does not remove duplicates?

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30. What is a subquery and where can it be used in a SQL statement?

> **Hint:** Think about SELECT, WHERE, and FROM clauses.

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31. Write a query to find employees whose salary is above the company average.

> **Hint:** Use a subquery to calculate the average.

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32. What is a Common Table Expression (CTE) and how is it created in SQL?

> **Hint:** Look for the WITH keyword.

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33. Give an example of when you would use a recursive CTE.

> **Hint:** Think about hierarchical data, like organizational charts.

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34. Explain what the OVER() clause does in window functions.

> **Hint:** It defines how the function is applied over rows.

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35. Write a query to show each employee's salary compared to the average salary in their department.

> **Hint:** Use AVG() OVER(PARTITION BY ...).

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36. What is the difference between ROW_NUMBER, RANK, and DENSE_RANK window functions?

> **Hint:** Focus on how they handle ties.

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37. Write a query to find the top 3 selling products in each region.

> **Hint:** Combine PARTITION BY and ROW_NUMBER.

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38. How would you extract the year and month from a date column in SQL?

> **Hint:** There are built-in functions for this.

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39. Write a query to calculate the number of days since a customer's last visit.

> **Hint:** Use a function that subtracts dates.

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40. How can you concatenate first and last name columns in SQL?

> **Hint:** There's a function for joining strings.

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41. Write a query to clean up extra spaces in a product code column.

> **Hint:** Look for a function that removes leading and trailing spaces.

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42. How would you classify customers into 'High', 'Medium', or 'Low' value segments using SQL?

> **Hint:** Use the CASE WHEN statement.

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43. What is a view in SQL and why would you use one for analytics reporting?

> **Hint:** Think about reusable queries and abstraction.

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44. How would you update a view if the underlying table structure changes?

> **Hint:** Consider the CREATE OR REPLACE statement.

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45. Why do queries sometimes run slowly, and how can indexing help?

> **Hint:** Consider how indexes affect data retrieval.

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46. How would you check the execution plan of a query in MySQL?

> **Hint:** There's a specific keyword for this.

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47. Describe the steps to import data from a CSV file into an SQL table.

> **Hint:** Think about the LOAD DATA command or GUI options.

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48. How can you connect Excel to a SQL database and use SQL queries within Excel?

> **Hint:** Look for built-in data connection features.

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49. Explain how Power Query in Excel can be used for dynamic data refresh from a SQL database.

> **Hint:** Focus on how queries can be parameterized and refreshed.

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50. What is the purpose of parameterized queries in SQL + Excel integration?

> **Hint:** Think about dynamic filtering and security.

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51. How would you use Python and pandas to run a SQL query and load the results into a DataFrame?

> **Hint:** Recall the function in pandas for SQL queries.

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52. Describe how you could combine SQL and Python for exploratory data analysis (EDA).

> **Hint:** Think about using SQL for extraction and Python for visualization.

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53. In a case study for Zomato Bangalore, how would you use SQL to identify top-rated restaurants by cuisine type?

> **Hint:** Consider grouping and sorting based on ratings and cuisine.

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54. Explain how SQL queries can help business teams make decisions about market segmentation using restaurant data.

> **Hint:** Think about grouping, filtering, and aggregating by customer or restaurant attributes.

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55. How could you use ChatGPT or another AI tool to help you write or debug a complex SQL query, and what steps would you take to ensure the AI-generated code is correct?

> **Hint:** Consider reviewing, testing, and understanding the output before using it.

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56. Give an example of how AI-powered analytics tools could be integrated with SQL databases to speed up data analysis, and where human judgment is still necessary.

> **Hint:** Think about data cleaning, pattern recognition, and interpreting results.
